Order Matters Beyond Accuracy: Paired Prediction Analysis of Compound Image Corruptions

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Repository & Data Artifacts: 973,400 forward passes on 3,925 Imagenette validation images across 2 regimes

WHAT THIS STUDY FINDS

1. Nominal-Order Benchmarking Conflates Sequence with Image Quality: Under standard ImageNet-C parameters, changing execution order across Gaussian Noise, Defocus Blur, and JPEG Compression induces top-1 accuracy disparities of up to 21.55% (ResNet-50) and 20.45% (ConvNeXt-Tiny) on identical physical images. These nominal differences are heavily driven by terminal operator smoothing: sequences ending in blur boost PSNR by ~6.2 dB and preserve accuracy, whereas sequences ending in noise or JPEG induce severe degradation.

2. Fidelity Calibration Contracts Aggregate Gaps but Leaves Residual Differences: Pre-calibrating parameters to target common (PSNR, SSIM) support contracts the cross-sequence range of the four-model mean by **63.18%** (from 18.43% down to 6.79%; architecture-specific gap contractions average **58.08%**, contracting by 77.15% in ConvNeXt, 72.62% in ResNet, 50.75% in ViT, and 31.80% in Swin). Prespecified tolerances (±0.75 dB PSNR, ±0.035 SSIM) were not achieved over the tested grid (PSNR RMSE 0.943 dB, SSIM RMSE 0.1380), establishing this strictly as an approximate-fidelity descriptive comparison rather than an exact causal effect at matched fidelity.

3. Order Matters Beyond Accuracy (Severe Error Migration): In a complete 5-state joint prediction partition (E_{11}, E_{10}, E_{01}, E_{00}^{same}, E_{00}^{diff}), **Predicted-Label Churn** (34.74%–40.78%) is more than double **Correctness Churn** (12.34%–16.93%) because Error Migration (E_{00}^{diff}) accounts for 22.40%–23.85% of all evaluations. Sequence pairs with near-identical aggregate accuracy (|ΔAcc| < 2.5%) exhibit up to 10.97% excess churn, proving that sequence order actively reshuffles decision boundaries across identical visual instances.

Abstract— Deep neural networks for computer vision are widely evaluated under synthetic distribution shifts, but standard benchmarks treat corruptions primarily as isolated unary operations (e.g., ImageNet-C) or along fixed unidirectional pipelines (e.g., R-Bench). In real-world imaging pipelines, degradations occur sequentially through multi-stage physical and digital processes governed by non-commutative operators ($T_A \blacksquare T_B \neq T_B \blacksquare T_A$). We conduct a systematic paired evaluation of the complete symmetric group of three canonical degradations—Gaussian Noise (N), Defocus Blur (B), and JPEG Compression (J)—yielding all 3! = 6 permutation sequences across 5 severity levels evaluated on four prominent vision architectures (ResNet-50, ConvNeXt-Tiny, ViT-Base/16, Swin-Tiny), comprising 973,400 total forward-pass evaluations across two distinct experimental regimes (486,700 records per regime on 3,925 FastAI Imagenette-v2 images with deterministically seed-matched noise realizations). We formalize two distinct estimands: (1) **Total Operational Effect** at standard ImageNet-C parameters, where operator non-commutativity drives cross-sequence top-1 accuracy gaps of up to 21.55% (ResNet-50) and 20.45% (ConvNeXt-Tiny) via downstream terminal smoothing; and (2) **Fidelity-Conditioned Comparison**, where pre-calibration targeting common (PSNR, SSIM) support contracts the cross-sequence range of the four-model mean accuracies by 63.18% (dropping from 18.43% down to 6.79%, while the average of the four individual architecture gap contractions is 58.08%, contracting by 77.15% in ConvNeXt and 72.62% in ResNet). Crucially, prespecified tolerances (±0.75 dB PSNR, ±0.035 SSIM) were not achieved (PSNR RMSE = 0.943 dB, SSIM RMSE = 0.1380), establishing this regime strictly as an approximate-fidelity descriptive comparison. Beyond aggregate accuracy, we formulate a complete 5-state joint prediction space partition (E_{11}, E_{10}, E_{01}, E_{00}^{same}, E_{00}^{diff}) and prove that **Predicted-Label Churn** ($P(Y\blacksquare_A \neq Y\blacksquare_B)$) is more than double **Correctness-State Churn** ($P(C_A \neq C_B) = L + G$), because Error Migration (E_{00}^{diff}) averages 23.85% in nominal pipelines and 22.40% under approximate fidelity. Pairs with near-identical accuracy (|ΔAcc| < 2.5%) exhibit up to 10.97% excess churn and over 40% label churn, proving that sequence order actively reshuffles classifier decision boundaries across identical visual instances.

1. Introduction & The Two Distinct Estimands

Benchmarking neural network robustness to distribution shift is essential for safety-critical deployment. While ImageNet-C (Hendrycks & Dietterich, 2019) established standard protocols for evaluating 15 isolated degradations, physical imaging pipelines subject images to multi-stage degradations. Because operators governing image formation (spatial convolution, stochastic noise, and block DCT quantization) are **non-commutative** ($T_A \blacksquare T_B \neq T_B \blacksquare T_A$), the sequence order in which operators act creates distinct physical signals and feature representations (Liao et al., 2020).

We explicitly formalize two distinct estimands:

• **Estimand 1 (Total Operational Effect at Fixed Parameters):** Evaluates the net impact of sequence order when operators run at identical standard nominal settings. Here, low-level fidelity disparities (e.g., higher PSNR when blur follows JPEG) are direct physical consequences of operator non-commutativity in real pipelines.

• **Estimand 2 (Fidelity-Conditioned Descriptive Comparison):** Evaluates whether cross-sequence accuracy disparities persist when operator parameters are calibrated to target common (PSNR, SSIM) support. Because discrete calibration did not satisfy prespecified tolerances (±0.75 dB PSNR, ±0.035 SSIM), this estimand provides a *descriptive comparison under approximate common support*, rather than an exact causal effect at matched fidelity. Confounding was reduced, but not eliminated.

2. Related Work & Novelty Demarcation

Prior benchmarks predominantly evaluate unary corruptions (ImageNet-C) or temporal video sequences (ImageNet-P). Dodge & Karam (2016, 2019) pioneered psychometric distortion matching on single corruptions. Goyal et al. (2022) studied dual corruptions (noise and motion blur in low-light imaging), noting order asymmetry; however, they evaluated only 2! = 2 orderings without calibrating parameters to common support across the full permutation group. Li et al. (2025) introduced R-Bench, evaluating a 7-stage physical pipeline on MLLMs, but followed a single unidirectional acquisition sequence without permuting stages. Liao et al. (2020) demonstrated in digital forensics that 2-operator chains leave detectable forensic traces, confirming operator non-commutativity. Wei et al. (2026) showed that paired single-image evaluations uncover classwise correctness bounds under unary visual perturbations. Prediction churn was formalized across model retraining iterations by Milani Fard, Cormier, Canini, & Gupta (2016). RobustLens provides the first full-factorial 3! = 6 permutation study on identical physical images with seed-matched noise, formulating a 5-state prediction space partition that decouples predicted-label churn from correctness churn, a full-rank 6-cell decomposition, and an empirical fidelity-calibration audit documenting gap contraction.

3. Mathematical Framework & Full 5-State Prediction Partition

Let x be a clean RGB image. Operators are Gaussian Noise (N_σ), Defocus Blur ($B_{\kappa,\gamma}$), and JPEG Compression (J_q). The symmetric group S_3 contains 6 sequences: NBJ, NJB, BNJ, BJN, JNB, JBN (applied left to right; e.g., NBJ ends in JPEG, NJB ends in Blur, BJN ends in Noise).

Let $\blacksquare_A$, $\blacksquare_B$ be predicted classes, Y ground truth, and C_A, $C_B \in \{0, 1\}$ correctness indicators. The joint prediction space partitions into five mutually exclusive, exhaustive transitions:

1. E_{11} (Both Correct): $\blacksquare_A = Y \wedge \blacksquare_B = Y$; 2. E_{10} (Loss / L): $\blacksquare_A = Y \wedge \blacksquare_B \neq Y$; 3. E_{01} (Gain / G): $\blacksquare_A \neq Y \wedge \blacksquare_B = Y$; 4. E_{00}^{same} (Same Wrong Class): $\blacksquare_A = \blacksquare_B \neq Y$; 5. E_{00}^{diff} (Error Migration / Different Wrong Classes): $\blacksquare_A \neq Y \wedge \blacksquare_B \neq Y \wedge \blacksquare_A \neq \blacksquare_B$.

Conservation of probability holds: $P(E_{11}) + P(E_{10}) + P(E_{01}) + P(E_{00}^{\text{same}}) + P(E_{00}^{\text{diff}}) = 1$.

We define and distinguish two churn metrics:

- **Correctness-State Churn:** $P(C_A \neq C_B) = P(E_{10}) + P(E_{01}) = L + G$
- **Predicted-Label Churn:** $P(\blacksquare_A \neq \blacksquare_B) = L + G + P(E_{00}^{\text{diff}})$ = Correctness Churn + Error Migration.
- **Agreement Rate:** $P(\blacksquare_A = \blacksquare_B) = P(E_{11}) + P(E_{00}^{\text{same}}) = 1$ - Label Churn.
- **Excess Churn:** Excess Correctness Churn = $2 \cdot \min(G, L) = (G+L) - |\Delta\text{Acc}|$; Excess Label Churn = $2 \cdot \min(G, L) + P(E_{00}^{\text{diff}})$.

Theorem: Whenever models make distinct incorrect predictions ($P(E_{00}^{\text{diff}}) > 0$), Correctness Churn strictly understates classifier instability. In our experiments, Error Migration is ~23%, causing Predicted-Label Churn to be more than double Correctness Churn.

4. Full-Rank Saturated Six-Cell Linear Model & Statistical Inference

To decompose sequence variance across all six cell means $\mu = [\mu_{NBJ}, \mu_{NJB}, \mu_{BNJ}, \mu_{BJN}, \mu_{JNB}, \mu_{JBN}]^T$, we specify the linear model $\mu = X\beta$ with reference levels Start Noise and End Noise:

Columns of X: [Intercept, Start Blur, Start JPEG, End Blur, End JPEG, Parity Contrast (Even: +1, Odd: -1)]. Because $\det(X) = -6 \neq 0$, X has full rank 6, uniquely determining $\beta = X^{-1}\mu$.

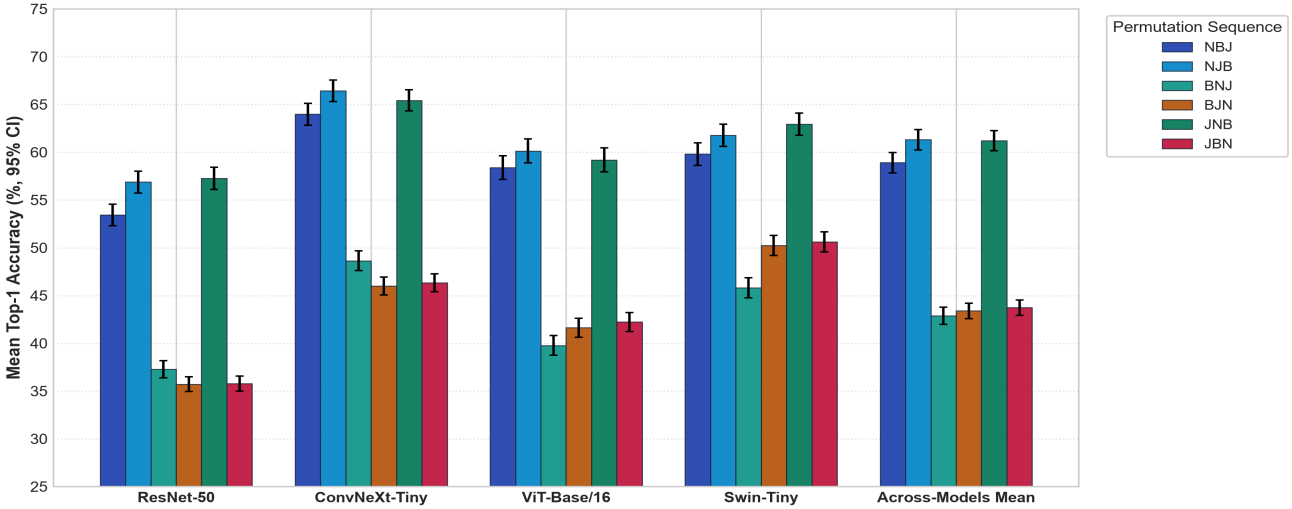
Critical Design Properties:

1. **Non-Orthogonal Design:** $X^T X \neq I$; reference-coded parameters must be interpreted alongside direct marginal contrasts.
2. **Unobserved Reference Condition:** No 3-operator sequence without repetition can both start and end with Noise. Hence, Intercept β_0 is an extrapolated linear parameter, not an observable baseline.
3. **Balanced Parity Contrast:** Even permutations (A_3: NBJ, BJN, JNB) and Odd permutations (NJB, BNJ, JBN) both contain every operator exactly once as an entry operator and exactly once as an exit operator. The coefficient $\beta_5 = (\mu\blacksquare_{\text{Even}} - \mu\blacksquare_{\text{Odd}})/2$ is perfectly balanced against entry and exit marginals, with full contrast $2\beta_5$.
4. **Clustered Bootstrap & Multiplicity:** We perform clustered bootstrap resampling ($B = 5{,}000$ clusters on 3,925 physical image IDs). Asymptotic two-sided Wald p-values are computed as $\log p = \log(2) + \text{logsf}(|z|)$ using cluster-bootstrap standard errors. Holm step-down adjustment controls FWER ($\alpha = 0.05$) on primary contrasts; Benjamini-Hochberg controls FDR ($q = 0.05$) on secondary families.

5. Regime 1: Nominal-Order Benchmark (Estimand 1)

Under fixed nominal parameters (ImageNet-C severities 1–5), models evaluated on identical physical images with matched noise seeds exhibit massive performance gaps.

Figure 1: Top-1 Classification Accuracy Across Six Nominal Corruption Permutations



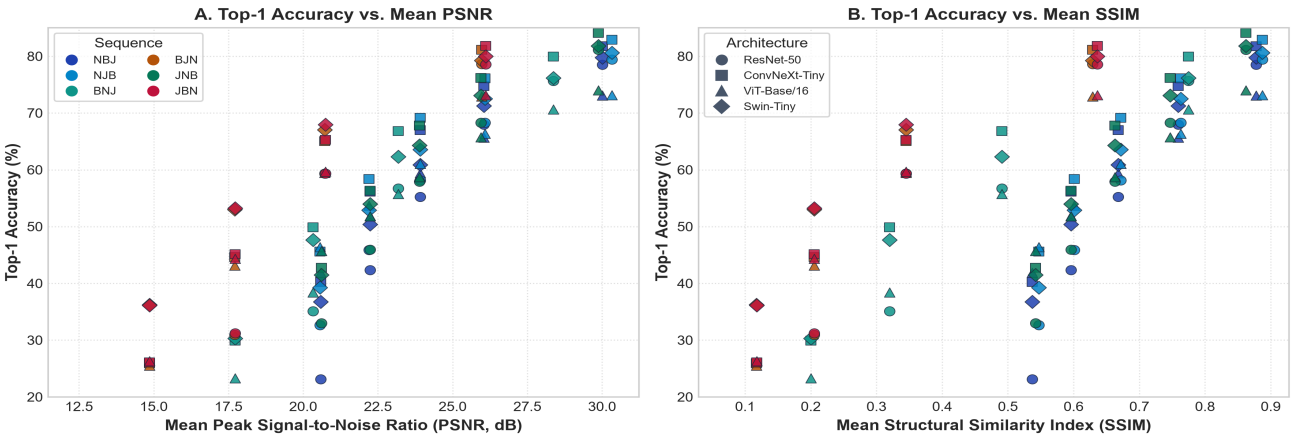
Model	Arch	NBJ	NJB	BNJ	BJN	JNB	JBN	Mean	Min	Max	Gap (Δ)	Brier
ResNet-50	CNN	53.44 %	56.88 %	37.30 %	35.71 %	57.27 %	35.79 %	46.06%	35.71 %	57.27 %	**21.55%**	0.1587
ConvNeXt-T	Modern CNN	63.98 %	66.44 %	48.63 %	45.99 %	65.44 %	46.33 %	56.14%	45.99 %	66.44 %	**20.45%**	0.1265
ViT-B/16	Transformer	58.39 %	60.14 %	39.77 %	41.64 %	59.19 %	42.23 %	50.23%	39.77 %	60.14 %	**20.37%**	0.1481
Swin-Tiny	Hier. ViT	59.81 %	61.78 %	45.81 %	50.24 %	62.94 %	50.63 %	55.20%	45.81 %	62.94 %	**17.13%**	0.1286
Overall Mean	—	58.90 %	61.31 %	42.88 %	43.40 %	61.21 %	43.75 %	51.91%	42.88 %	61.31 %	**18.43%**	0.1405

Table 1: Nominal-Order Top-1 Accuracy & Brier Scores. Evaluated across 3,925 images (19,625 evaluations per sequence per model; 471,000 compound records).

Direct Marginal Contrasts: Start Blur vs. Noise: -16.97% ($p < 10^{-100}$); Start JPEG vs. Noise: -7.63% ($p < 10^{-100}$). End Blur vs. Noise: **+17.69%** ($p < 10^{-100}$); End JPEG vs. Noise: +7.32% ($p < 10^{-100}$). Balanced Parity Contrast: **+5.19%** ($2\beta_5$, $p < 10^{-100}$).

Finding: In nominal evaluations, the massive 18.43% sequence gap is dominated by terminal operator identity: ending in Blur confers a +17.69% marginal boost over terminal noise because spatial convolution attenuates prior high-frequency noise and DCT block artifacts, boosting PSNR by ~6.2 dB (mean PSNR 24.6 dB for terminal blur vs. 18.3 dB for terminal noise).

Figure 2: Accuracy Broadly Tracks Image Fidelity Across Nominal Orders



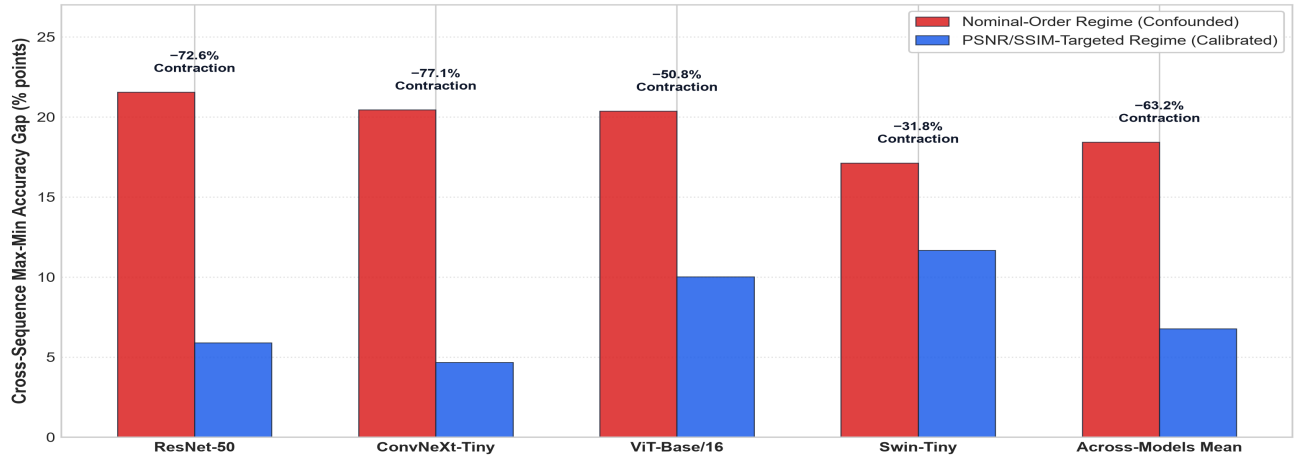
6. Regime 2: PSNR/SSIM-Targeted Descriptive Comparison (Estimand 2)

To test whether order sensitivity persists when output distortion is constrained, operator parameters were pre-calibrated on an independent training partition (N_{cal} = 50) targeting five common (PSNR, SSIM) support bins across sequences.

Target	Target PSNR	Target SSIM	Max Abs PSNR Resid	PSNR Tol (≤ 0.75 dB)	Max Abs SSIM Resid	SSIM Tol (≤ 0.035)
Target 1	28.0 dB	0.850	1.117 dB (NJB)	**Failed**	0.0686 (BJN)	**Failed**
Target 2	24.5 dB	0.740	0.936 dB (BJN)	**Failed**	0.1758 (BJN)	**Failed**
Target 3	21.5 dB	0.620	1.260 dB (NBJ)	**Failed**	0.2379 (BJN)	**Failed**
Target 4	18.5 dB	0.480	1.989 dB (NBJ)	**Failed**	0.2373 (BJN)	**Failed**
Target 5	16.0 dB	0.360	2.639 dB (NBJ)	**Failed**	0.2067 (BJN)	**Failed**
Overall	—	—	**2.639 dB (Max)**	**RMSE: 0.943 dB (Failed)**	**0.2379 (Max)**	**RMSE: 0.1380 (Failed)**

Model	NBJ	NJB	BNJ	BJN	JNB	JBN	Matched Mean	Matched Gap	Nominal Gap	Contraction	Mean Brier
ResNet-50	50.15 %	46.76 %	45.18 %	50.63 %	47.05 %	51.08 %	48.47%	**5.90%**	21.55%	**72.62%**	0.1558
ConvNeXt-T	60.65 %	56.46 %	55.97 %	59.22 %	56.47 %	59.86 %	58.11%	**4.67%**	20.45%	**77.15%**	0.1245
ViT-B/16	56.54 %	53.39 %	46.51 %	54.72 %	53.01 %	55.68 %	53.31%	**10.03%**	20.37%	**50.75%**	0.1444
Swin-Tiny	57.61 %	50.44 %	53.93 %	61.23 %	52.31 %	62.12 %	56.27%	**11.68%**	17.13%	**31.80%**	0.1433
Overall Mean	56.23 %	51.76 %	50.40 %	56.45 %	52.21 %	57.18 %	54.04%	**6.79%**	18.43%	**63.18%**	0.1420

Figure 3: Contraction of Sequence Accuracy Gaps in PSNR/SSIM-Targeted Regime



Parameter	Nominal Estimate (%)	Nominal 95% Clustered CI	Nominal p_{Holm}	Matched Estimate (%)	Matched 95% Clustered CI	Matched p_{Holm}
β_0 **(Intercept)**	+51.63	[+50.71, +52.52]	$< 10^{-100}$	+59.51	[+58.51, +60.51]	$< 10^{-100}$
β_1 **(Start Blur)**	-10.83	[-11.14, -10.53]	$< 10^{-100}$	-3.99	[-4.22, -3.75]	$< 10^{-100}$
β_2 **(Start JPEG)**	-5.29	[-5.50, -5.08]	$< 10^{-100}$	-1.40	[-1.58, -1.23]	$< 10^{-50}$
β_3 **(End Blur)**	**+12.27**	[+11.88, +12.67]	$< 10^{-100}$	**-6.82**	[-7.18, -6.46]	$< 10^{-100}$
β_4 **(End JPEG)**	**+4.67**	[+4.35, +5.00]	$< 10^{-100}$	**-4.20**	[-4.50, -3.91]	$< 10^{-100}$
β_5 **(Parity Contrast)**	**+2.60**	[+2.52, +2.67]	$< 10^{-100}$	**+0.93**	[+0.86, +0.99]	$< 10^{-100}$

Table 4: Saturated Six-Cell Decomposition Parameters. Even after conditioning on entry and exit operators, the balanced parity contrast β_5 remains statistically significant in both regimes (+2.60% nominal, +0.93% matched, $p < 10^{-100}$).

7. Prediction Instability, Error Migration & Per-Class Breakdown

A cornerstone empirical finding is that **aggregate accuracy obscures massive per-image prediction churn and error migration.**

Figure 4: 5-State Prediction Space Partition & Decoupling of Label Churn from Correctness Churn



Regime	Model	Both Corr (E_{11})	Loss (E_{10})	Gain (E_{01})	Same Wrong (E_{00}^{same})	Error Mig (E_{00}^{diff})	Corr Churn	Excess Churn	Label Churn	Agree Rate	Mean |ΔConf|
Nominal	ConvNeXt	47.63%	11.64%	5.38%	14.99%	**20.36%**	17.02%	5.36%	**37.38%**	62.62%	0.1662
Nominal	ResNet-50	36.96%	12.06%	6.15%	14.90%	**29.93%**	18.20%	5.72%	**48.14%**	51.86%	0.1850
Nominal	Swin-Tiny	47.98%	8.49%	5.96%	16.06%	**21.51%**	14.45%	5.82%	**35.96%**	64.04%	0.1835
Nominal	ViT-Base	41.20%	11.76%	6.31%	17.17%	**23.57%**	18.06%	6.68%	**41.63%**	58.37%	0.1732
Nominal	**Overall**	**43.44%**	**10.99%**	**5.95%**	**15.78%**	**23.85%**	**16.93%**	**5.90%**	**40.78%**	**59.22%**	**0.1770**
Matched	ConvNeXt	52.64%	5.49%	5.44%	17.44%	**19.00%**	10.93%	8.51%	**29.92%**	70.08%	0.1163
Matched	ResNet-50	41.92%	6.18%	6.92%	15.54%	**29.44%**	13.10%	10.15%	**42.54%**	57.46%	0.1481
Matched	Swin-Tiny	50.31%	4.78%	7.14%	18.03%	**19.74%**	11.92%	5.99%	**31.66%**	68.34%	0.1307
Matched	ViT-Base	46.61%	6.61%	6.79%	18.55%	**21.44%**	13.40%	9.44%	**34.85%**	65.15%	0.1314
Matched	**Overall**	**47.87%**	**5.76%**	**6.57%**	**17.39%**	**22.40%**	**12.34%**	**8.52%**	**34.74%**	**65.26%**	**0.1316**

Table 5: Complete 5-State Joint Prediction Space Transitions. Error Migration (E_{00}^{diff}) accounts for 22.40%–23.85% of all evaluations, pushing Predicted-Label Churn to more than double Correctness Churn.

Class Name	Nom Acc	Nom Conf	Nom Brier	Nom Label Churn	Nom Corr Churn	Nom Error Mig	Matched Acc	Matched Label Churn	Matched Error Mig
tench	73.57%	66.25%	0.0921	29.09%	17.56%	11.54%	77.22%	21.54%	11.91%
golf ball	72.45%	66.65%	0.1088	28.86%	17.89%	10.97%	75.07%	21.88%	10.83%
parachute	68.01%	61.06%	0.1224	31.19%	16.22%	14.97%	68.97%	28.00%	15.75%
chain saw	55.56%	51.44%	0.1170	41.93%	17.42%	24.51%	59.32%	34.59%	22.64%
French horn	51.98%	52.58%	0.1053	41.77%	14.21%	27.56%	53.81%	36.00%	27.08%
garbage truck	49.61%	50.66%	0.1528	43.23%	19.56%	23.67%	51.25%	36.10%	21.24%
gas pump	48.57%	48.14%	0.1253	46.52%	17.51%	29.01%	50.56%	40.01%	28.27%
cassette player	41.35%	42.38%	0.1988	45.62%	20.64%	24.98%	44.40%	39.95%	22.40%
English springer	40.48%	40.22%	0.1786	48.72%	17.85%	30.87%	41.61%	42.85%	28.25%
church	**18.35%**	38.21%	0.2065	**50.45%**	**11.04%**	**39.41%**	19.22%	**46.07%**	**34.70%**

8. Scope Boundaries, Limitations & Reproducibility

1. **Approximate Fidelity, Not Exact Matching:** Prespecified tolerances (±0.75 dB PSNR, ±0.035 SSIM) were not achieved over the tested discrete grid. Residual PSNR spread reached 3.10 dB at Target 5. Residual differences cannot be interpreted as pure architectural effects without acknowledging residual distortion imbalance.
2. **Finite Grid Boundary:** The failure of our discrete parameter search does not prove universal physical impossibility across continuous spaces or alternate corruption formulations.
3. **Dataset & Task Scope:** Evaluated on FastAI Imagenette-v2 (10 classes, 3,925 images). Evaluations on full ImageNet-1k and dense prediction tasks remain open for future work.
4. **Data Availability:** All 973,400 raw prediction records and reproducible scripts are archived in the repository.

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